

Program 4: Integrate git with docker for source-controlled application builds

File structure:

```
/Dockerfile  
/package.json  
/package-lock.json  
/index.js
```

Create folder and initialize package.json

```
mkdir program4 > cd program4 > npm init -y > npm i express
```

Dockerfile

#stage 1

FROM node:18 AS build

WORKDIR /app

COPY package*.json ./

RUN npm install

COPY . .

#stage 2

FROM node:18-slim

WORKDIR /app

COPY --from=build /app /app

EXPOSE 3000

CMD ["node","index.js"]

index.js

```
const express = require('express')
```

```
const app = express()
```

```
app.get('/',(req,res)=>res.send("Hello from 3rd program"))
```

```
app.listen(3000,()=>{console.log("Server running on http://localhost:3000");})
```

Run the following command to create docker image and container

```
sudo docker build -t program4 .
```

```
sudo docker run -d -p 3000:3000 --name pro4 program4
```

```
1RV24MC036_gayatri_v_k@ubuntu:~$ mkdir program4
1RV24MC036_gayatri_v_k@ubuntu:~$ cd program4
1RV24MC036_gayatri_v_k@ubuntu:~/program4$ npm init -y
Wrote to /home/1RV24MC036_GAYATRI_V_K/program4/package.json:
```

```
{
  "name": "program4",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}
```

```
1RV24MC036_gayatri_v_k@ubuntu:~/program4$ nano index.js
1RV24MC036_gayatri_v_k@ubuntu:~/program4$ npm install express
```

added 68 packages, and audited 69 packages in 2s

18 packages are looking for funding
run `npm fund` for details

found 0 vulnerabilities

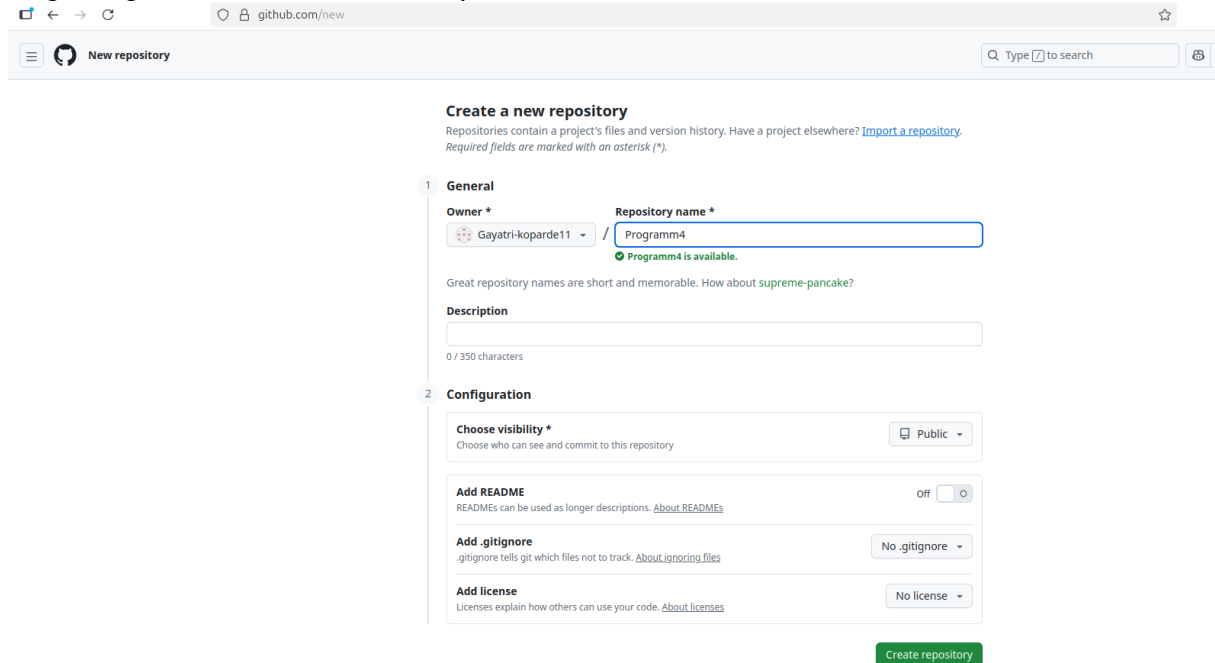
```
GNU nano 7.2
const express = require("express");
const app = express();
const PORT = 3000;

app.get("/", (req, res) => {
  res.send("Hello from Git + Docker Integration!");
});

app.listen(PORT, () => {
  console.log(`Server running at http://localhost:${PORT}`);
});
```

Go to browser and type <http://localhost:3000> to see the output

Login to github account create a repo



Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1 **General**

Owner * Gayatri-koparde11 / **Repository name *** Programm4
✔ Programm4 is available.

Great repository names are short and memorable. How about supreme-pancake?

Description
0 / 350 characters

2 **Configuration**

Choose visibility * Choose who can see and commit to this repository. Public

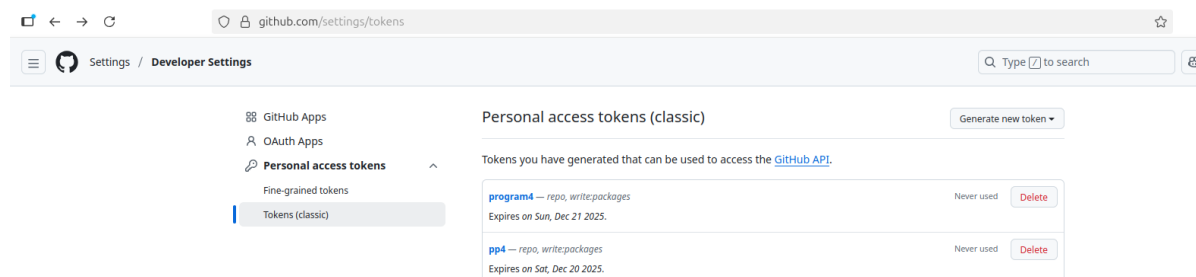
Add README READMEs can be used as longer descriptions. [About READMEs](#). Off

Add .gitignore .gitignore tells git which files not to track. [About ignoring files](#). No .gitignore

Add license Licenses explain how others can use your code. [About licenses](#). No license

Create repository

Go to profile > settings > developer settings > access tokens > general tokens > create token > copy the token password and keep it safe



github.com/settings/tokens

Settings / Developer Settings

Personal access tokens (classic) Generate new token

Tokens you have generated that can be used to access the [GitHub API](#).

program4 — repo, write:packages Expires on Sun, Dec 21 2025.	Never used	Delete
pp4 — repo, write:packages Expires on Sat, Dec 20 2025.	Never used	Delete

Initialize your directory with git

git init

git add .

git commit -m "<message>"

git branch -M main

Add remote git repo to cli

git remote add origin https://github.com/<username>/<repo>.git

git push -u origin main

```
1RV24MC036_gayatri_v_k@ubuntu:~/program4$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/1RV24MC036_GAYATRI_V_K/program4/.git/
1RV24MC036_gayatri_v_k@ubuntu:~/program4$ git add .
1RV24MC036_gayatri_v_k@ubuntu:~/program4$ git commit -m "Initial commit"
[master (root-commit) 2044687] Initial commit
636 files changed, 72751 insertions(+)
create mode 100644 Dockerfile
create mode 100644 index.js
create mode 100644 node_modules/.package-lock.json
create mode 100644 node_modules/accepts/HISTORY.md
create mode 100644 node_modules/accepts/LICENSE
create mode 100644 node_modules/accepts/README.md
create mode 100644 node_modules/accepts/index.js
create mode 100644 node_modules/accepts/package.json
create mode 100644 node_modules/body-parser/HISTORY.md
create mode 100644 node_modules/body-parser/LICENSE
create mode 100644 node_modules/body-parser/README.md
create mode 100644 node_modules/body-parser/index.js
create mode 100644 node_modules/body-parser/lib/read.js
create mode 100644 node_modules/body-parser/lib/types/json.js
create mode 100644 node_modules/body-parser/lib/types/raw.js
create mode 100644 node_modules/body-parser/lib/types/text.js
create mode 100644 node_modules/body-parser/lib/types/urleencoded.js
create mode 100644 node_modules/body-parser/lib/utils.js
create mode 100644 node_modules/body-parser/package.json
create mode 100644 node_modules/bytes/HISTORY.md
create mode 100644 node_modules/bytes/LICENSE
create mode 100644 node_modules/bytes/Readme.md
create mode 100644 node_modules/bytes/index.js
create mode 100644 node_modules/bytes/package.json
create mode 100644 node_modules/call-bind-apply-helpers/.eslintrc
create mode 100644 node_modules/call-bind-apply-helpers/.github/FUNDING.yml
create mode 100644 node_modules/call-bind-apply-helpers/.nycrc
create mode 100644 node_modules/call-bind-apply-helpers/CHANGELOG.md
```

Go to repo and see the files were added

The screenshot shows a GitHub repository page for a user named 'Gayatri-koparde11' with the repository name 'program4'. The page is public. The repository was created 33 minutes ago and contains 1 commit. The files listed are: node_modules, Dockerfile, index.js, package-lock.json, and package.json. The README section is empty, with a prompt to 'Add a README'. The right sidebar shows the repository has 0 stars, 0 forks, and 0 releases. The language statistics show JavaScript at 53.4% and Dockerfile at 46.6%.

Gayatri-koparde11 / program4

main 1 Branch 0 Tags

Go to file Add file Code

Gayatri-koparde11 Initial commit 2044687 · 33 minutes ago 1 Commit

node_modules	Initial commit	33 minutes ago
Dockerfile	Initial commit	33 minutes ago
index.js	Initial commit	33 minutes ago
package-lock.json	Initial commit	33 minutes ago
package.json	Initial commit	33 minutes ago

README

Add a README

Help people interested in this repository understand your project by adding a README.

[Add a README](#)

About

No description, website, or topics provided.

Activity

0 stars

0 watching

0 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Languages

JavaScript 53.4% Dockerfile 46.6%

To list and remove container and images

sudo docker container ls -a

sudo docker container stop <container id>

sudo docker container rm <container id>

sudo docker images

sudo docker rmi <imageid>